

Lecture 1: Memory, Storage and Peripherals

RAM (DRAM/SRAM) · ROM/EPROM/EEPROM · Cache · Virtual Memory · HDD · SSD · Optical · USB · SD Cards

Learning Outcomes

- Distinguish between memory and secondary storage, and describe the memory hierarchy from registers down to secondary storage.
- Explain how RAM works, and differentiate between DRAM and SRAM in terms of construction, speed, cost, and use.
- Describe ROM and trace the evolution of its family: Mask ROM, PROM, EPROM, and EEPROM.
- Explain cache memory, the principle of locality, and the difference between a cache hit and a cache miss.
- Explain virtual memory, paging, page faults, and the problem of thrashing.
- Describe the internal working and performance factors of an HDD, and the working of NAND flash inside an SSD.
- Compare HDD, SSD, optical storage (CD/DVD/Blu-ray), USB flash drives, and SD cards, and choose the right storage for a given need.
- Answer short, medium, and long-answer (up to 15 marks) examination questions on this topic with full explanations, diagrams, and examples.

1 Memory vs Storage — and the Memory Hierarchy

Every computer, phone, or game console needs two very different kinds of “remembering,” and the distinction between them is one of the most important ideas in this lecture.

Point	Memory (Primary)	Secondary Storage
What it holds	The instructions and data a running program needs right now	Files, programs, and the operating system, permanently
Speed	Extremely fast, so the CPU is never kept waiting	Much slower to access than memory
Volatility	Usually volatile — contents vanish when power is lost	Non-volatile — data survives even with the power off
Everyday example	The space actively in use while you work	Every photo, document, and installed app you own

The Memory Hierarchy

Main Objectives of a Memory Hierarchy:

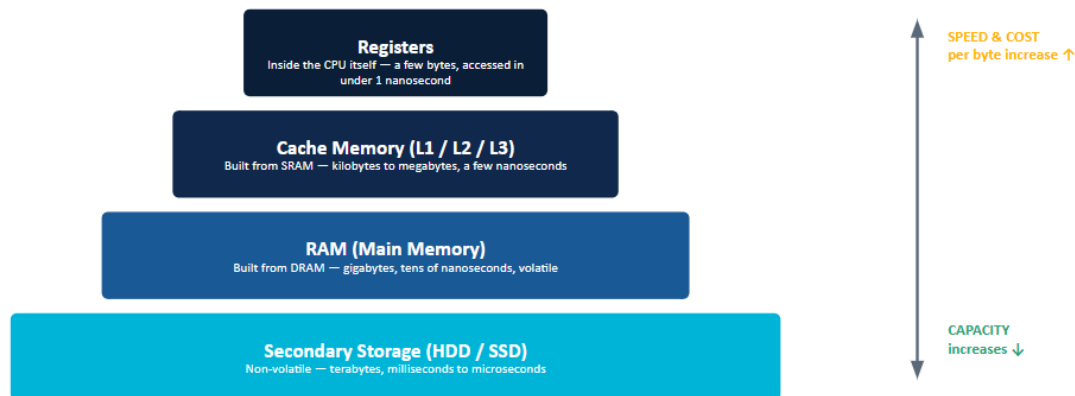
1. To minimize execution time of executing programs
2. To maximize the throughput of the computer
3. To minimize response time

subject to the constraint that high-speed memory is limited in size.

Every computer layers several kinds of storage on top of one another. The higher a layer sits, the faster — and more expensive per byte — it is; the lower a layer sits, the bigger and cheaper per byte it becomes.

The Memory Hierarchy

Every computer layers several kinds of storage — the higher the layer, the faster and more expensive it is



Golden rule: the CPU always checks the fastest, closest layer first (registers, then cache), and only moves down to slower layers (RAM, then disk) when the data it needs is not already there.

Fig 1.1 — The memory hierarchy: speed and cost per byte decrease left to right; capacity increases left to right.

Golden Rule for Exams

The CPU always checks the fastest, closest layer first (registers, then cache) and only moves down to slower layers (RAM, then disk) when the data it needs is not already there. This single idea explains why cache memory and virtual memory both exist — you will meet both later in this lecture.

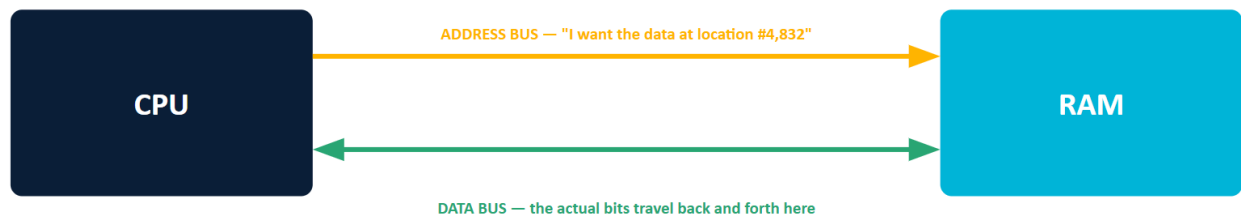
2 RAM — Random Access Memory

RAM is the computer's main working memory. Whenever a program — a browser, a game, a word processor — is opened, the operating system copies its instructions and current data from storage into RAM, because the CPU can read and write RAM thousands of times faster than it can read a disk.

Property	Explanation
Very Fast	Typical access time is 10–50 nanoseconds — billions of operations per second.
Randomly Addressable	Any memory cell can be reached directly using its address, in constant time, unlike a tape that must be read sequentially.
Volatile	All contents are lost the instant power is removed — which is why unsaved work disappears after a crash or shutdown.

How the CPU Talks to RAM

Every access between the CPU and RAM happens over two separate sets of wires: the address bus and the data bus.



1. The CPU places the memory address it needs on the address bus ("I want the data at location #4,832").
2. RAM decodes that address and locates the exact row and column of memory cells.
3. Data flows along the data bus — into the CPU on a read, or out of it on a write.
4. Because addressing is direct, this whole exchange takes only tens of nanoseconds, regardless of where in RAM the data is stored.

3 DRAM and SRAM — The Two Kinds of RAM

DRAM — Dynamic RAM

DRAM is the technology used for the main memory sticks inside almost every PC, laptop, and phone. Each bit of DRAM is stored as an electrical charge in a tiny capacitor, paired with one transistor that acts as a switch to read or write that charge — a "1 transistor + 1 capacitor" (1T1C) design. Because this design is extremely small, DRAM chips can pack billions of bits into a tiny area, making it cheap enough for large capacities.

Why it needs "refreshing": a capacitor slowly leaks its charge, like a bucket with a tiny hole — within milliseconds the stored bit would fade and be lost. To prevent this, the memory controller reads and rewrites (refreshes) every single cell thousands of times per second. This refresh cycle uses power and briefly blocks normal access, which is part of why DRAM is slower than SRAM.

DRAM Generation	Approx. Speed	Notes
DDR3	~1600 MT/s	Low power; found in older laptops and desktops
DDR4	~3200 MT/s	Most common generation in 2015–2022 PCs
DDR5	~4800+ MT/s	Current generation; higher bandwidth

DRAM Generation	Approx. Speed	Notes
What Does “DDR” Mean? DDR stands for Double Data Rate: data is transferred on both the rising and the falling edge of the clock signal, which doubles throughput compared to transferring data on just one edge.		

SRAM — Static RAM

SRAM is the technology used for CPU cache and registers, where every nanosecond counts. Each bit of SRAM is stored in a small circuit called a flip-flop, typically built from six transistors. A flip-flop is bistable — it naturally settles into one of two stable states (0 or 1) and stays there as long as power is supplied, with no leakage to compensate for.

Property	Explanation
No Refresh Needed	Because the flip-flop holds its state actively, there is no leakage to correct — SRAM is ready to be accessed on every single clock cycle.
Much Faster	Access times are often under 1–2 nanoseconds — fast enough to keep up with a modern CPU core running at several gigahertz.
Larger & Costlier	Six transistors per bit (versus DRAM's two components) means far fewer bits fit in the same silicon area, so SRAM is reserved for small, critical memory like cache.

DRAM vs SRAM — Side by Side

Attribute	DRAM	SRAM
Storage element	1 capacitor + 1 transistor	6-transistor flip-flop
Needs refreshing?	Yes, thousands of times/sec	No
Speed	Tens of nanoseconds	Under 1–2 nanoseconds
Cost per bit	Low	High (5–10× more)

Attribute	DRAM	SRAM
Density	Very high (GB per chip)	Low (KB–MB per chip)
Power usage	Higher (refresh cycles)	Lower when idle
Typical use	Main system RAM	CPU cache & registers
Important Point for Exams Both DRAM and SRAM are volatile — both lose their contents completely when power is switched off. “Volatile” is not unique to DRAM; it is the correct answer for RAM in general, regardless of type.		

4 ROM and Its Family — Mask ROM, PROM, EPROM, EEPROM

ROM (Read Only Memory) is a non-volatile memory chip — its contents survive completely intact even with the power off. It is used to store fixed instructions the computer needs immediately at start-up, before it can even load an operating system from disk, such as the BIOS or UEFI firmware, as well as firmware in devices like routers, printers, and washing machines.

Point	RAM	ROM
Volatility	Volatile — loses data when powered off	Non-volatile — keeps data with no power at all
Read/Write	Read AND write — changes constantly during use	Read-only, or very hard to change
What it stores	Currently running programs and data	Permanent start-up and firmware code

Point	RAM	ROM
Typical capacity	Large, measured in gigabytes	Small, measured in kilobytes to megabytes

The ROM Family — A Story of Increasing Flexibility

ROM is not a single technology — it is a family, and each new member solved the previous one's biggest limitation: how to change what's stored.

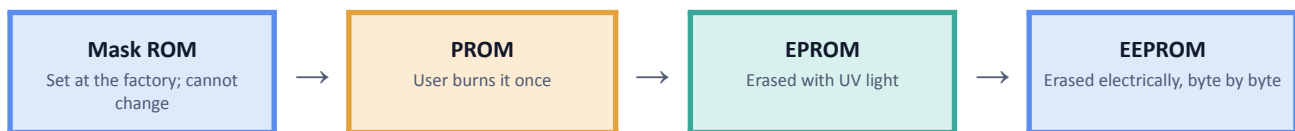


Fig 1.2 — The ROM family: each generation made stored data progressively easier to change.

Mask ROM

The data pattern is built directly into the chip's physical circuitry during manufacturing, using a photographic “mask.” It is extremely cheap to produce in bulk, but only economical for very large orders of the exact same chip — and it cannot be changed after manufacturing, so a single design mistake means the whole batch is scrapped.

PROM (Programmable Read-Only Memory)

PROM is sold blank, then written once by the customer using a special device called a “PROM programmer” or “burner.” Inside, each bit is a tiny fuse; programming sends a pulse of current that permanently blows selected fuses — an intact fuse reads as one value, a blown fuse reads as the other. Once fuses are blown, the change is irreversible. PROM was historically used in simple calculators, toys, and early game cartridges, and remains useful for small production runs where Mask ROM's setup cost is not worthwhile.

EPROM (Erasable PROM)

EPROM stores each bit in a “floating-gate transistor” — a transistor with an extra, electrically isolated gate that can trap electrons. Once trapped during programming, the electrons stay put for years because the floating gate is surrounded by an insulator with no path to leak away, which is what gives EPROM its non-volatility.

How it's erased: older EPROM chips have a small quartz window on top of the package. Shining strong ultraviolet (UV) light through this window for 10–20 minutes gives the trapped electrons enough energy to escape, resetting every cell at once — there is no way to erase a single byte, only the whole chip together, which made EPROM impractical for frequent updates.

EEPROM (Electrically Erasable PROM)

EEPROM uses the same floating-gate principle as EPROM, but adds a thinner insulating layer that allows electrons to tunnel back out under an applied electric field (a phenomenon called Fowler-Nordheim tunneling). Erasing no longer needs UV light: it can be done electrically, right inside the running device, and often one byte at a time instead of the whole chip at once.

Limited write cycles: each erase-write cycle slightly wears the insulating layer; typical EEPROM chips are rated for around 100,000 to 1,000,000 write cycles per cell. EEPROM is also the ancestor

of Flash memory (used in SSDs, USB drives, and SD cards), which is essentially EEPROM redesigned to erase in large blocks at once, trading fine-grained control for much higher speed and density.

The ROM Family — Full Comparison

Type	Written By	Erase Method	Erase Granularity
Mask ROM	Manufacturer only	Cannot be erased	N/A
PROM	User, once, with a burner	Cannot be erased	N/A
EPROM	User, with a programmer	UV light (10–20 min)	Entire chip only
EEPROM	User, in-circuit	Electrical signal	Byte by byte
Flash (descendant)	User, in-circuit	Electrical signal	Whole block at a time

5 Cache Memory

Modern CPUs can execute an instruction in under a nanosecond, but fetching data from RAM takes tens of nanoseconds — a huge mismatch that would leave the CPU idle most of the time. Cache is a small pool of very fast SRAM sitting between the CPU and RAM that keeps a copy of recently and frequently used data close by, so the CPU rarely has to wait for slower RAM.

The Principle of Locality — Why Caching Works

Type of Locality	Explanation
Temporal Locality	If a piece of data was used recently, it is very likely to be used again soon — for example, a loop counter variable that is read and updated on every pass through a loop.
Spatial Locality	If one memory address was used, nearby addresses are also likely to be used soon — for example, the next few instructions in a program, or adjacent elements of an array.

Cache Levels — L1, L2 & L3

Cache is organised as a tiered system: each level trades some speed for much greater size. The farther a level sits from the CPU core, the larger and slower it is — but all three levels remain dramatically faster than a trip out to RAM.

Level	Typical Size	Typical Speed	Shared?
L1	32–64 KB per core	~1 ns	No — private per core

Level	Typical Size	Typical Speed	Shared?
L2	256 KB–2 MB per core	~3–10 ns	Usually private per core
L3	8–32 MB total	~10–20 ns	Yes — shared across all cores

Cache Hit vs Cache Miss

Outcome	What Happens	Typical Figure
Cache Hit	The data the CPU needs is already sitting in cache; it is delivered in just a few nanoseconds — the fastest possible outcome, with no waiting.	Typical hit rate: 90–97%
Cache Miss	The needed data is not in cache. The CPU must wait while it is fetched from the slower RAM, and that data (plus its neighbours, thanks to spatial locality) is then copied into cache for next time.	Costs 10–20× longer than a hit

Outcome	What Happens	Typical Figure
Why Hit Rate Matters So Much Even a small drop in hit rate has an outsized effect on performance, because a miss costs 10–20 times longer than a hit. This is why CPU designers spend enormous effort on cache size, associativity, and prefetching algorithms that predict what data will be needed next.		

6 Virtual Memory

Virtual memory is a technique, built into the operating system and CPU together, that gives every running program the illusion of having a large, private, continuous block of memory all to itself — even though physical RAM is smaller and shared among many programs at once. Behind the scenes, the operating system can move data that is not currently needed out of RAM and onto disk, freeing up space for what is needed right now.

Benefit	Explanation
Isolation	Each program gets its own private address space, so one program cannot accidentally read or corrupt another program's memory.
Larger Apparent Memory	Programs can be written as if far more RAM exists than is physically installed.
Simplified Programming	Every program can assume it starts at the same clean address space, without needing to know what else is running.

How Paging Works

Memory is split into equal-sized blocks called pages, tracked by a page table that looks up where each page actually lives right now — in physical RAM, or parked out on disk.

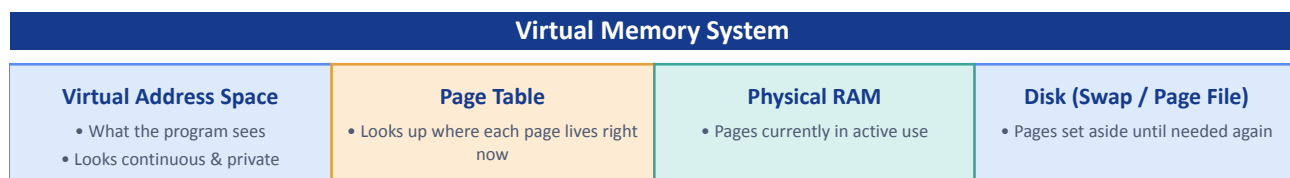


Fig 1.3 — Paging: the page table maps each virtual page to either physical RAM or disk.

What Is a Page Fault?

A page fault happens when the CPU asks for a page that the page table shows is out on disk, not in RAM. The operating system pauses that program, loads the page back into RAM (possibly moving a different, less-used page out to make room), updates the page table, and then lets the program continue — all invisibly to the program itself.

Virtual Memory — Trade-offs

The Benefits	The Trade-off
Lets you run more programs at once than physical RAM would otherwise allow.	Disk is far slower than RAM, so every page fault causes a noticeable delay.
Protects programs from interfering with each other's memory.	Excess swapping consumes disk bandwidth that other tasks also need.
Makes large programs possible even on modest hardware.	On mechanical hard disks especially, this can make the whole system feel sluggish.

The Benefits	The Trade-off
Thrashing — When Virtual Memory Backfires If too many programs demand more memory than physically exists, the operating system ends up constantly swapping pages in and out — barely finishing one page fault before triggering the next. The CPU spends almost all of its time managing memory instead of doing real work, and the system can slow to a crawl even though it is “busy.” This condition is called thrashing, and the usual fix is simply closing programs or adding more physical RAM.	
7 Memory Types — Full Recap	

Type	Volatile?	Speed	Typical Size	Main Use
SRAM (Cache)	Yes	Fastest (~1 ns)	KB–MB	Speeds up CPU access
DRAM (RAM)	Yes	Fast (tens of ns)	GB	Runs active programs
ROM / Mask ROM	No	Slower	KB–MB	Fixed factory instructions
PROM	No	Slower	KB–MB	One-time custom programming
EPROM	No	Slower	KB–MB	UV-reprogrammable firmware
EEPROM	No	Slower	KB–MB	Electrically-updatable firmware
Virtual Memory	No*	Slowest	GB (uses disk)	Extends RAM using disk space

Type	Volatile?	Speed	Typical Size	Main Use
Note *Virtual memory itself is a technique, not a physical chip — it borrows non-volatile disk space to make the system behave as if it had more RAM.				

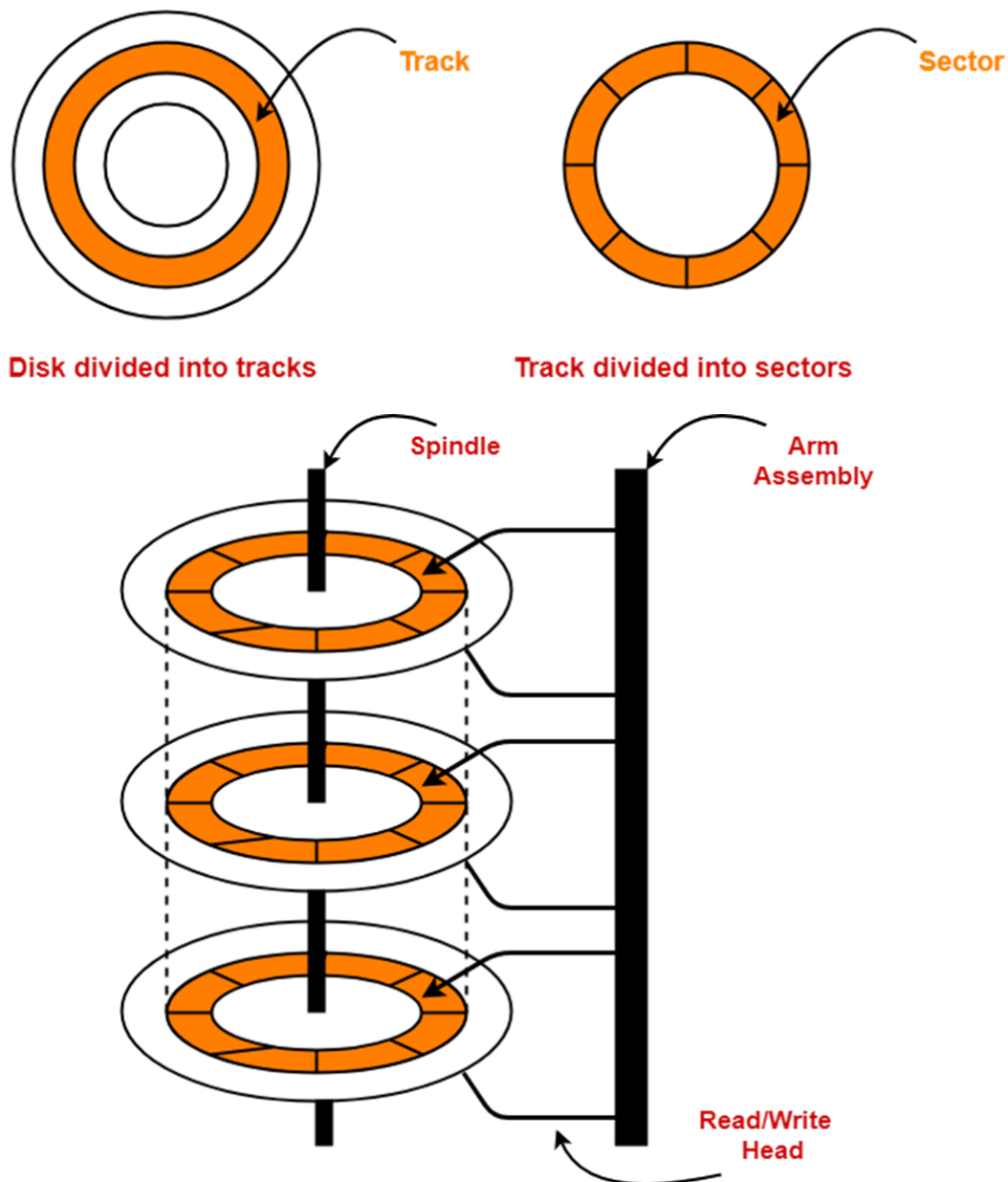
8 Secondary Storage: HDD

A Hard Disk Drive (HDD) is a precisely engineered stack of spinning magnetic disks. It is a thin plastic or metallic circular plate coated with magnetic oxide and encased in a protective cover. Data is stored on magnetic disks as magnetized spots.

- The entire disk is divided into platters.
- Each platter consists of concentric circles called as tracks.
- These tracks are further divided into sectors which are the smallest divisions in the disk, historically 512 bytes and now commonly 4,096 bytes.

Hard Disk Drive (HDD)			
Platter • Rigid magnetic disk storing data	Spindle Motor • Spins platters at 5,400–7,200 RPM	Read/Write Head • Floats nm above platter, senses charges	Actuator Arm • Sweeps the head across the platter

Fig 1.4 — The internal components of an HDD.



HDD — Understanding Performance

Because parts must physically move, HDD speed is described by several distinct measurements.

Measurement	What It Means	Typical Figure
Seek Time	How long the actuator arm takes to physically move the head to the correct track.	~4–10 ms
Rotational Latency	The extra wait for the platter to spin around until the right sector reaches the head.	~2–4 ms
Data Transfer Rate	How fast data streams off the platter once the head is correctly positioned.	80–160 MB/s
Spindle Speed	How fast the platters spin — higher RPM generally means lower latency and faster transfer.	5,400–7,200 RPM

Measurement	What It Means	Typical Figure
Important Point for Exams Seek time + rotational latency together are called “access time” — this is the reason random reads (jumping around the disk) are so much slower than sequential reads (reading one long stretch in order).		

•Consider a hard disk with:

4 surfaces

64 tracks/surface

128 sectors/track

256 bytes/sector

Q:1 What is the capacity of the hard disk?

$$\text{Disk capacity} = \text{surfaces} * \text{tracks/surface} * \text{sectors/track} * \text{bytes/sector}$$

$$\text{Disk capacity} = 4 * 64 * 128 * 256$$

$$\text{Disk capacity} = 8 \text{ MB}$$

•Consider a disk pack with the following specifications- 16 surfaces, 128 tracks per surface, 256 sectors per track and 512 bytes per sector.

Q: What is the capacity of disk pack?

Capacity of disk pack = Total number of surfaces x Number of tracks per surface x Number of sectors per track x Number of bytes per sector

= 16 x 128 x 256 x 512 bytes

= 228 bytes

= 256 MB

9 Secondary Storage: SSD

A Solid State Drive (SSD) is built from the same floating-gate transistor idea as EEPROM, organised for speed instead of fine-grained control.

SSD — How Flash Memory Works

Built from the same floating-gate transistor idea as EEPROM, organized for speed



NAND Flash Memory Chips

Cells, Pages & Blocks

Each transistor cell stores one or more bits. Cells are grouped into pages (typically 4–16 KB, the smallest unit that can be written), and pages are grouped into blocks (typically 128–256 pages, the smallest unit that can be erased).

Why You Can't Overwrite Directly

A NAND cell can only be written after it has been erased, and erasing works on a whole block at once. So updating a single page usually means writing the new data to a fresh, already-erased page and marking the old one invalid — handled automatically by the controller.

The Controller's Job

A dedicated controller chip manages wear leveling, error correction, and the mapping between the addresses the computer sees and the physical cells actually used — all invisibly and in real time.

NAND Flash Memory Chip		
Cell One transistor; stores 1+ bits	Page 4–16 KB; smallest unit that can be WRITTEN	Block 128–256 pages; smallest unit that can be ERASED

Fig 1.5 — Cells, pages, and blocks inside NAND flash memory.

Why you can't overwrite directly: a NAND cell can only be written after it has been erased, and erasing works on a whole block at once. So updating a single page usually means writing the new data to a fresh, already-erased page and marking the old one invalid — handled automatically by the controller.

The controller's job: a dedicated controller chip manages wear levelling, error correction, and the mapping between the addresses the computer sees and the physical cells actually used — all invisibly and in real time.

SSD — NAND Types & Key Technology

NAND Type	Bits per Cell	Endurance	Cost	Typical Use
SLC	1 bit	Highest (~100,000 cycles)	Very high	Enterprise & industrial
MLC	2 bits	High (~10,000 cycles)	High	Performance drives
TLC	3 bits	Moderate (~3,000 cycles)	Moderate	Most consumer SSDs
QLC	4 bits	Lower (~1,000 cycles)	Lowest	Budget, high-capacity drives
Technique	Purpose			
Wear Levelling	The SSD controller spreads writes evenly across all blocks so no single block wears out early, extending the drive's overall lifespan.			

NAND Type	Bits per Cell	Endurance	Cost	Typical Use
TRIM Command	The operating system tells the SSD which blocks hold deleted data, so the controller can erase them in advance instead of waiting until the space is needed — keeping write speeds high over time.			

HDD vs SSD — Full Comparison

Factor	HDD	SSD
Underlying tech	Spinning magnetic platters	NAND flash memory chips
Sequential speed	80–160 MB/s	500–7,000+ MB/s
Random access	Slow (physical seek needed)	Very fast (electronic addressing)
Moving parts	Yes — motor, platters, arm	None
Noise & vibration	Audible, some vibration	Completely silent
Shock resistance	Sensitive to drops/bumps	Highly shock-resistant
Power draw	Higher (spinning motor)	Lower, especially idle
Price per GB	Lowest	Higher, but falling steadily
Common interface	SATA	SATA or NVMe (PCIe)
Best for	Bulk/archival storage	OS, applications, gaming

Factor	HDD	SSD
Common Real-World Approach Pair a smaller SSD (for the operating system and frequently used apps) with a larger HDD (for bulk file storage) to balance speed and cost.		

10 Optical Storage — CD, DVD & Blu-ray

Underneath a protective clear coating, an optical disc has a spiral track of microscopic bumps, called “pits,” separated by flat areas called “lands,” pressed or burned into a reflective metal layer. A laser shines onto this spinning track and a sensor measures the reflected light: a land reflects strongly, while a pit scatters the light and reflects weakly. The pattern of strong and weak reflections as the disc spins is decoded into the 1s and 0s of digital data.

What Increases Capacity	How
Smaller Laser Wavelength	A shorter wavelength laser can be focused into a tighter spot, allowing smaller pits to be read reliably — the single biggest reason newer formats hold more data.
Multiple Layers	Discs can stack two or more semi-transparent reflective layers, letting the laser refocus to read a second layer through the first — roughly doubling capacity.
Track Density	Newer formats also pack the spiral track itself more tightly, fitting more total track length — and therefore more data — onto the same-sized disc.

CD vs DVD vs Blu-ray — Technical Specs

Spec	CD	DVD	Blu-ray
Laser colour / wavelength	Infrared, 780 nm	Red, 650 nm	Blue-violet, 405 nm
Single-layer capacity	~700 MB	4.7 GB	25 GB
Dual-layer capacity	N/A	8.5 GB	50 GB (up to 100+ GB, 3–4 layers)
Typical use	Music, small data	SD video, software	HD/4K video, large backups
Introduced	1982	1995	2006

Important Point for Exams

The pattern to remember:
shorter laser wavelength → smaller focusable spot → smaller pits → more data packed into the same physical disc size. All three formats are also available as recordable (CD-R/DVD-R/BD-R, written once) or rewritable (CD-RW/DVD-RW/BD-RE, erasable and reusable) variants, which use a heat-sensitive dye layer instead of pressed pits.

11 USB Flash Drives & USB Standards

A USB flash drive combines a small NAND flash chip, a controller, and a USB connector in one compact package — the same NAND flash technology as an SSD, packaged for portability. Plugging

it into any USB port lets it draw power and transfer data with no extra software; the drive appears instantly as a removable disk on Windows, macOS, or Linux.

USB Standard	Also Called	Max Theoretical Speed	Approx. Era
USB 2.0	Hi-Speed USB	480 Mbps (~60 MB/s)	2000–2010
USB 3.0	SuperSpeed / USB 3.1 Gen 1	5 Gbps (~625 MB/s)	2010–2015
USB 3.1 Gen 2	SuperSpeed+	10 Gbps (~1,250 MB/s)	2015–2018
USB 3.2 / USB4	SuperSpeed 20Gbps+	20–40 Gbps	2018–present

Note: real-world speeds are usually lower than the theoretical maximum, limited by the flash chip inside the drive, not just the USB port.

Common uses: transferring files between computers, running portable applications without installing them, creating a bootable OS installer or recovery drive, and quick backups of important documents.

12 SD Cards — Types & Speed Classes

SD Cards — Types & Speed Classes

Physical Sizes



Capacity Classes

SD / SDSC	up to 2 GB	File system: FAT16
SDHC	4 GB – 32 GB	File system: FAT32
SDXC	64 GB – 2 TB	File system: exFAT

Speed Classes — Why They Matter

The rating printed on a card guarantees its minimum sustained write speed — critical for continuous 4K video recording or burst-mode photography, where a slow card causes dropped frames or missed shots.

Class 2/4/6/10	UHS Speed Class U1 / U3	Video Speed Class V30/V60/V90	App Performance A1/A2
min 2–10 MB/s	min 10 / 30 MB/s	min 30/60/90 MB/s	random I/O for apps

Physical Sizes

Format	Notes
SD (Standard)	32 × 24 mm
miniSD	Legacy format, now rare

Format	Notes
microSD	15 × 11 mm; used in phones and drones

Capacity Classes

Class	Capacity Range	File System
SD / SDSC	up to 2 GB	FAT16
SDHC	4 GB – 32 GB	FAT32
SDXC	64 GB – 2 TB	exFAT

Speed Classes — Why They Matter

The rating printed on a card guarantees its minimum sustained write speed — critical for continuous 4K video recording or burst-mode photography, where a slow card causes dropped frames or missed shots.

Speed Class System	Minimum Guaranteed Speed
Class 2 / 4 / 6 / 10	2–10 MB/s
UHS Speed Class U1 / U3	10 / 30 MB/s
Video Speed Class V30 / V60 / V90	30 / 60 / 90 MB/s
App Performance A1 / A2	Rated for random I/O for apps, not just sequential speed

13 Choosing the Right Storage

Approximate maximum common capacities, from smallest to largest: CD (~0.7 GB) → DVD (~8.5 GB) → Blu-ray (~100 GB) → USB drive (up to ~1 TB) → SD card (up to ~2 TB) → SSD (up to ~4 TB+) → HDD (up to ~20 TB+). Real products vary by model and price point.

If You Need...	Choose...
Your PC to boot and load apps fast	An SSD
Cheap, huge storage for backups or media	An HDD
To carry files between computers	A USB Flash Drive
To shoot photos or video on a camera or drone	An SD Card (check its Speed Class)
To share or archive movies or software offline	Blu-ray or DVD

If You Need...	Choose...
14 Putting It All Together	

Consider what happens when you open a large photo-editing project you saved yesterday and start working on it:

Step	Component Involved	What Happens
1	SSD / HDD	The project file is read from secondary storage, where it was saved permanently.
2	RAM	The operating system copies the file's data into RAM, since the CPU can read/write RAM thousands of times faster than disk.
3	Cache (L1/L2/L3)	As you apply an edit, the CPU keeps pulling the same pixel data and instructions into cache, exploiting temporal and spatial locality.
4	Virtual Memory	If you open many large images at once and RAM fills up, the OS pages out less-used data to disk, freeing RAM — you might notice a brief slowdown (a page fault).
5	SSD / HDD (again)	When you hit Save, the edited file is written back to secondary storage, permanently, surviving the next power-off.

This single everyday task touches almost every idea in this lecture: the speed gap between RAM and storage, why cache exists, how virtual memory can both help and slow things down, and why volatile memory needs non-volatile storage behind it.

15 Fill in the Blanks

- 1. RAM that stores each bit as a charge in a capacitor is called _____ RAM.
- 2. _____ RAM uses a six-transistor flip-flop and needs no refreshing.
- 3. The CPU sends a memory address to RAM along the _____ bus, and the actual bits travel along the _____ bus.
- 4. ROM that is written once by the user with a special “burner” device is called _____.
- 5. _____ is erased using ultraviolet light shone through a small quartz window.
- 6. _____ can be erased and rewritten electrically, one byte at a time, without removing it from the circuit.
- 7. The tendency to reuse the same data very soon is called _____ locality; the tendency to use nearby addresses soon is called _____ locality.
- 8. When the CPU requests data that is not currently in cache, this is called a cache _____.
- 9. A _____ occurs when the CPU requests a page that has been swapped out to disk.
- 10. In an HDD, the time taken for the platter to rotate until the correct sector reaches the head is called _____.

Answers

1. Dynamic (DRAM) | 2. Static (SRAM) | 3. address, data | 4. PROM | 5. EPROM | 6. EEPROM | 7. temporal, spatial | 8. miss | 9. page fault | 10. rotational latency

16 True or False

- 1. SRAM needs to be refreshed thousands of times per second.
- 2. ROM is volatile memory.
- 3. EEPROM can be erased byte by byte, electrically, without removing it from the circuit.
- 4. Cache memory is built from DRAM.
- 5. A cache hit is slower than a cache miss.
- 6. Virtual memory allows a computer to run more programs than physical RAM would otherwise allow.
- 7. An SSD has moving mechanical parts like an HDD.
- 8. TLC NAND stores three bits per cell.

Answers with Explanation

1. False. It is the reverse — DRAM needs constant refreshing because its capacitor leaks charge; SRAM's flip-flop holds its state actively and needs no refresh.

2. False. ROM is non-volatile — it keeps its data even without power, which is exactly why it is used for start-up firmware.

3. True. That in-circuit, byte-level electrical erase is precisely what distinguishes EEPROM from EPROM, which needs UV light and erases the whole chip at once.

4. False. Cache is built from SRAM, not DRAM, because SRAM is dramatically faster — essential for keeping up with the CPU.

5. False. A cache hit is much faster than a cache miss; a miss costs roughly 10–20 times longer because the data must be fetched from slower RAM.

6. True. Virtual memory extends apparent memory using disk space via paging, letting more programs run than physical RAM alone would allow.

7. False. An SSD has no moving parts at all — that is one of its biggest advantages over an HDD, which relies on a spinning platter and moving head.

8. True. TLC stands for Triple-Level Cell, storing 3 bits in every cell.

17 Short Answer Questions (2–3 Marks)

- Differentiate between memory and secondary storage. (3 marks)
- Name the four levels of the memory hierarchy in order, from fastest to slowest. (2 marks)
- Define DRAM and SRAM. (2 marks)
- What is the key difference between EPROM and EEPROM? (3 marks)
- What is cache memory, and why is it needed? (2 marks)
- Define virtual memory and page fault. (3 marks)
- List the four main internal parts of an HDD. (2 marks)

- Differentiate between an HDD and an SSD. (3 marks)
- Name the three main optical storage formats with one typical use of each. (2 marks)

18 Long Answer Model Answers (5, 10 & 15 Marks)

The following are fully written model answers. Use these as a template for structure, depth, and the level of explanation expected in your own exam answer — do not simply memorise them word for word; practise writing them in your own words using the same structure.

Q1 (5 Marks): Differentiate between DRAM and SRAM, and explain why each is used where it is.

DRAM (Dynamic RAM) stores each bit as an electrical charge in a tiny capacitor paired with one transistor. Because a capacitor slowly leaks charge, DRAM must be refreshed thousands of times per second, which makes it somewhat slower, but this 1-transistor design is extremely small and cheap, allowing billions of bits to be packed into one chip. This is why DRAM is used as a computer's main RAM, where large capacity at low cost matters most.

SRAM (Static RAM) stores each bit in a six-transistor flip-flop, which is bistable and holds its state actively without leaking, so it needs no refreshing and can be accessed in under 1–2 nanoseconds. However, six transistors per bit make SRAM far larger and 5–10 times more costly per bit than DRAM. This is why SRAM is reserved for CPU cache and registers — small amounts of memory where speed matters far more than cost or capacity.

Q2 (10 Marks): Explain the ROM family — Mask ROM, PROM, EPROM, and EEPROM — describing how each is written and erased, and how the family evolved.

ROM (Read Only Memory) is non-volatile memory used to store fixed instructions a computer needs at start-up. Over time, four related technologies developed, each solving the previous one's biggest weakness: the difficulty of changing what is stored.

Type	Written By	Erase Method	Erase Granularity
Mask ROM	Manufacturer only, during production	Cannot be erased	N/A
PROM	User, once, using a burner device	Cannot be erased	N/A
EPROM	User, with a programmer	UV light for 10–20 minutes	Entire chip only
EEPROM	User, in-circuit	Electrical signal	Byte by byte

Mask ROM is the least flexible: its contents are set permanently into the silicon during manufacturing, making it cheap in bulk but impossible to change afterwards. PROM improved on this slightly by letting the user “burn” the chip once themselves, blowing internal fuses that cannot be unblown. EPROM went further by making the chip reusable: shining UV light through a small quartz window resets the entire chip's floating-gate transistors at once, allowing full reprogramming, though not selectively. EEPROM solved this final limitation by using a thinner insulating layer that allows an applied electric field to erase electrically, right inside the running device, one byte at a time — without UV light or removing the chip.

In conclusion, each stage of the ROM family reduced the effort needed to change stored data, from “impossible after manufacturing” (Mask ROM), to “one irreversible factory-like pass” (PROM), to “UV light and 20 minutes for the whole chip” (EPROM), to finally “an electrical pulse in milliseconds, byte by byte” (EEPROM) — and EEPROM's block-erase descendant, Flash memory, is what now powers SSDs, USB drives, and SD cards.

Q3 (15 Marks): “The memory hierarchy is a compromise between speed, cost, and capacity.” Discuss this statement with reference to registers, cache, RAM, and secondary storage, and explain how cache memory and virtual memory each help bridge the gap between a fast CPU and slower storage.

This statement is entirely accurate. No single memory technology can be simultaneously the fastest, the cheapest per byte, and the largest in capacity — so every computer instead layers several different technologies, each making a different trade-off.



(A labelled memory hierarchy diagram, as in Fig 1.1 of these notes, should be drawn here in your answer sheet to earn full diagram marks.)

Registers sit inside the CPU itself, holding only a few bytes but accessed in under a nanosecond — the fastest and most expensive tier. **Cache**, built from SRAM, holds kilobytes to megabytes and is still measured in a few nanoseconds. **RAM**, built from DRAM, holds gigabytes at tens of nanoseconds, but is volatile. **Secondary storage** (HDD/SSD) holds terabytes non-volatily, but at millisecond-to-microsecond speeds — thousands of times slower than RAM.

Cache memory bridges the CPU-to-RAM gap: since a CPU can execute an instruction in under a nanosecond but RAM takes tens of nanoseconds to respond, cache exploits temporal and spatial locality to keep a copy of recently and nearby-used data close to the CPU, so that most requests are served from cache (a “hit”) instead of waiting on RAM (a “miss,” which costs 10–20 times longer).

Virtual memory bridges the RAM-to-disk gap in the opposite direction: rather than speeding up access, it extends RAM's apparent capacity by “borrowing” disk space through paging, moving pages that are not currently needed out to disk to make room. This lets more programs run than physical RAM alone would allow, at the cost of occasional page faults — and, if overused, the severe slowdown known as thrashing.

In conclusion, the memory hierarchy exists precisely because speed, cost, and capacity cannot be maximised all at once. Cache memory manages the CPU-RAM speed gap by keeping likely-needed data close by, while virtual memory manages RAM's limited capacity by borrowing space from slower but far larger secondary storage — together allowing a computer to feel both fast and capable of running far more than its physical RAM alone could hold.

Q4 (10 Marks): Compare HDD, SSD, and optical storage in terms of technology, speed, and typical use, and explain how you would choose between HDD, SSD, USB flash drive, SD card, and optical disc for different real-world needs.

HDDs, SSDs, and optical discs each store data non-volatily, but using very different technology, which leads to very different performance and typical uses.

Storage Type	Underlying Technology	Typical Speed	Typical Use
HDD	Spinning magnetic platters, read by a moving head	80–160 MB/s; slow random access	Cheap, bulk/archival storage
SSD	NAND flash memory chips, no moving parts	500–7,000+ MB/s; fast random access	Operating system, applications, gaming
Optical (CD/DVD/Blu-ray)	Laser-read pits and lands on a spinning disc	Slowest of the three; varies by format	Offline distribution/archiving of media and software

Choosing between them depends entirely on the need at hand. An SSD is the right choice when a computer needs to boot and load applications quickly, since its electronic addressing avoids the seek time and rotational latency that slow an HDD down. An HDD remains the better choice for cheap, high-capacity backup or media storage, where raw speed matters less than cost per gigabyte. A USB flash drive is ideal for carrying files between computers or creating a bootable installer, since it is small, portable, and needs no extra software. An SD card is the natural choice for a camera or drone, provided its Speed Class is high enough to sustain continuous 4K video recording without dropped frames. Finally, an optical disc such as a Blu-ray or DVD remains useful for sharing or archiving movies and software offline, particularly where long-term, write-once permanence is valued over speed.

In conclusion, no single storage type is best for every situation; real systems commonly combine several — for example, pairing a fast SSD for the operating system with a large HDD for bulk storage — choosing each technology for the specific trade-off of speed, capacity, cost, and portability it offers.

19 Exam Tips

- Memory trick for the hierarchy: “Registers, Cache, RAM, Storage” (“R-C-R-S”) — speed and cost fall as you go down the list; capacity rises.
- Memory trick for the ROM family: “Mask → PROM → EPROM → EEPROM” — each step makes erasing easier: never, once, UV light, electrically.
- For “differentiate” questions (DRAM vs SRAM, HDD vs SSD, EPROM vs EEPROM), always answer using a table — it is faster to write and easier for the examiner to award marks clearly.
- Always draw the relevant diagram (memory hierarchy, ROM family, or HDD/NAND components) whenever a question asks you to “explain” or “describe” these topics — diagrams carry dedicated marks in most university marking schemes.
- Common Mistake: Do not say cache is “built from RAM” — cache is built from SRAM specifically, which is a different technology from the DRAM used for main RAM.
- Common Mistake: Do not confuse a cache miss with a page fault — a miss means data was not in cache (so it is fetched from RAM); a page fault means data was not in RAM at all (so it must be fetched from disk).
- For 10 and 15 mark questions, structure your answer with a clear paragraph or table for each component, exactly like the model answers in Section 18 — examiners reward clear structure.

Glossary of Key Terms

Term	Meaning
Volatile	Loses its stored data when power is removed.
Non-volatile	Keeps its stored data even without power.
Latency	The delay between requesting data and receiving the first byte of it.
Bandwidth / Throughput	How much data can be transferred per second once a transfer is underway.
Firmware	Low-level software permanently stored on a chip that controls a device's basic operation.
Bus	A shared set of wires that carries addresses, data, or control signals between components.
Page fault	An event where the CPU requests data that has been swapped out to disk and must be reloaded.
NAND flash	A type of non-volatile memory, descended from EEPROM, used in SSDs, USB drives, and SD cards.
Wear levelling	A technique that spreads writes evenly across flash memory to prevent early failure.
Sector	The smallest addressable unit of data on a disk, historically 512 bytes.

MDC Section II, Lecture 1 — Memory, Storage and Peripherals — the foundation for later lectures on operating systems and file management.